

Use of Immunomodulators and Biologics Before, During, and After Pregnancy.

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Immunomodulators and biologic medications, alone or in combination, form the core therapeutic strategy for managing moderate-to-severe inflammatory bowel disease (IBD). IBD incidence peaks during the prime reproductive years, raising concerns about the impact of disease and its treatment on fertility, maternal and fetal health during pregnancy, breastfeeding safety, and childhood development. Although IBD increases risk of pregnancy complications independent of disease activity, adverse pregnancy outcomes are more common when disease is active. To mitigate fetal risk, women should conceive while disease is quiescent. Aside from methotrexate, immunomodulators and biologics may be used during pregnancy to achieve and maintain disease control. Based on available safety data, there is no increased risk of congenital anomalies among infants exposed to these medications. Active thiopurine metabolites and most monoclonal antibodies cross the placenta and are detectable in neonates. They are detectable in breast milk in minute levels as well. The impact of this exposure on neonatal outcomes is discussed. Adjusted dosing schedules during gestation may reduce fetal drug exposure, though the maternal risks of such manipulation require careful consideration. Ongoing prospective studies will further inform risk assessment, including for newer medications such as the anti-integrin agents.

Ebola Outbreak Response in the DRC with rVSV-ZEBOV-GP Ring Vaccination.

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At the beginning of the 2018-2020 outbreak of Ebola virus disease (EVD) in eastern Democratic Republic of Congo (DRC), no vaccine had been licensed. However, cluster-randomized evidence from Guinea in 2015 had indicated that ring vaccination around new cases (targeting contacts and contacts-of-contacts) with the use of single-dose live-replicating rVSV-ZEBOV-GP vaccine reduced EVD rates starting 10 days after vaccination. Thus, ring vaccination was added to the standard control measures for that outbreak. In this study, we evaluated the incidence of EVD within the first 9 days after vaccination (when little protection was expected from case isolation or ring vaccination), during days 10 to 29, and at later time periods. We established 1853 rings around new cases or clusters within 21 days after symptom onset in the index case and offered vaccination to the ring members. Vaccinees were monitored for EVD onset until the end of the outbreak in mid-2020. From August 8, 2018, to January 14, 2020, we vaccinated 265,183 participants. Of these vaccinees, 102,515 were monitored on days 0, 3, and 21 for safety. Among the contacts and contacts-of-contacts, 434 cases of EVD (0.2 per ring) were diagnosed, almost all within 0 to 9 days (380 cases) or 10 to 29 days (32 cases) after vaccination. An additional 22 cases were diagnosed after day 29 during an average of 170 more days of follow-up. The sooner that control measures (including ring vaccination) began after EVD onset in the index case, the sooner EVD rates fell among contacts. In each subgroup, EVD rates fell suddenly around day 10. Among the contacts and contacts-of-contacts who were still disease-free at day 10, the EVD onset rate during days 10 to 29 was 0.16 per 1000 (in 32 of 194,019 participants). This rate was much lower than the rate of 4.64 per 1000 (in 21 of 4528 participants) that had been seen among similarly defined ring members in Guinea, in whom standard control measures had been promptly initiated but vaccination was delayed until 21 days after ring formation (rate ratio, 0.04; 95% confidence interval, 0.02 to 0.06). No safety concerns with the vaccine were identified. Nonrandomized evidence regarding standard EVD control measures plus ring vaccination in eastern DRC reinforces the earlier randomized evidence from Guinea of vaccine efficacy against EVD onset 10 or more days after vaccination.

Multimerization of Ebola GPΔmucin on protein nanoparticle vaccines has minimal effect on elicitation of neutralizing antibodies.

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Ebola virus (EBOV), a member of the Filoviridae family of viruses and a causative agent of Ebola Virus Disease (EVD), is a highly pathogenic virus that has caused over twenty outbreaks in Central and West Africa since its formal discovery in 1976. The only FDA-licensed vaccine against Ebola virus, rVSV-ZEBOV-GP (Ervebo®), is efficacious against infection following just one dose. However, since this vaccine contains a replicating virus, it requires ultra-low temperature storage which imparts considerable logistical challenges for distribution and access. Additional vaccine candidates could provide expanded protection to mitigate current and future outbreaks. Here, we designed and characterized two multimeric protein nanoparticle subunit vaccines displaying 8 or 20 copies of GPΔmucin, a truncated form of the EBOV surface protein GP. Single-dose immunization of mice with GPΔmucin nanoparticles revealed that neutralizing antibody levels were roughly equivalent to those observed in mice immunized with non-multimerized GPΔmucin trimers. These results suggest that some protein subunit antigens do not elicit enhanced antibody responses when displayed on multivalent scaffolds and can inform next-generation design of stable Ebola virus vaccine candidates.

Recurrent Ebolavirus disease in the Democratic Republic of Congo: update and challenges.

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The current Ebolavirus disease (EVD) outbreak in the provinces of North Kivu and Ituri is the tenth outbreak affecting the Democratic Republic of Congo (DRC); the first outbreak occurring in a war context, and the second most deadly Ebolavirus outbreak on record following the 2014 outbreak in West Africa. The DRC government's response consisted of applying a package of interventions including detection and rapid isolation of cases, contact tracing, population mapping, and identification of high-risk areas to inform a coordinated effort. The coordinated effort was to screen, ring vaccinate, and conduct laboratory diagnoses using GeneXpert (Cepheid) polymerase chain reaction. The effort also included ensuring safe and dignified burials and promoting risk communication, community engagement, and social mobilization. Following the adoption of the "Monitored Emergency Use of Unregistered Products Protocol," a randomized controlled trial of four investigational treatments (mAb114, ZMapp, and REGN-EB3 and Remdesivir) was carried out with all consenting patients with laboratory-confirmed EVD. REGN-EB3 and mAb114 showed promise as treatments for EVD. In addition, one investigational vaccine (rVSV-ZEBOV-GP) was used first, followed by a second prophylactic vaccine (Ad26.ZEBOV/MVA-BN-Filo) to reinforce the prevention. Although the provision of clinical supportive care remains the cornerstone of EVD outbreak management, the DRC response faced daunting challenges including general insecurity, violence and community resistance, appalling poverty, and entrenched distrust of authority. Ebolavirus remains a public health threat. A fully curative treatment is unlikely to be a game-changer given the settings of transmission, zoonotic nature, limits of effectiveness of any therapeutic intervention, and timing of presentation.

Dose-dependent T-cell Dynamics and Cytokine Cascade Following rVSV-ZEBOV Immunization.

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The recent West African Ebola epidemic led to accelerated efforts to test Ebola vaccine candidates. As part of the World Health Organisation-led VSV Ebola Consortium (VEBCON), we performed a phase I clinical trial investigating rVSV-ZEBOV (a recombinant vesicular stomatitis virus-vectored Ebola vaccine), which has recently demonstrated protection from Ebola virus disease (EVD) in phase III clinical trials and is currently in advanced stages of licensing. So far, correlates of immune protection are incompletely understood and the role of cell-mediated immune responses has not been comprehensively investigated to date. We recruited 30 healthy subjects aged 18-55 into an open-label, dose-escalation phase I trial testing three doses of rVSV-ZEBOV (3×10^5 plaque-forming units (PFU), 3×10^6 PFU, 2×10^7 PFU) (ClinicalTrials.gov; NCT02283099). Main study objectives were safety and immunogenicity, while exploratory objectives included lymphocyte dynamics, cell-mediated immunity and cytokine networks, which were assessed using flow cytometry, ELISpot and LUMINEX assay. Immunization with rVSV-ZEBOV was well tolerated without serious vaccine-related adverse events. Ebola virus-specific neutralizing antibodies were induced in nearly all individuals. Additionally, vaccinees, particularly within the highest dose cohort, generated Ebola glycoprotein (GP)-specific T cells and initiated a cascade of signaling molecules following stimulation of peripheral blood mononuclear cells with Ebola GP peptides. In addition to a benign safety and robust humoral immunogenicity profile, subjects immunized with 2×10^7 PFU elicited higher cellular immune responses and stronger interlocked cytokine networks compared to lower dose groups. To our knowledge these data represent the first detailed cell-mediated immuneprofile of a clinical trial testing rVSV-ZEBOV, which is of particular interest in light of its potential upcoming licensure as the first Ebola vaccine. VEBCON trial Hamburg, Germany (NCT02283099).

Durability of single-dose rVSV-ZEBOV vaccine responses: what do we know?

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The live-attenuated recombinant vesicular stomatitis virus vaccine expressing the glycoprotein (GP) of Zaire Ebola virus (rVSV-ZEBOV) has proven immunogenic in humans and effective in field studies. Yet long-term durability of vaccine responses is unknown. We survey the evidence available in the literature for the durability of human responses to rVSV-ZEBOV. We also review determinants of initial responses and of their persistence. Persistence of EBOV-GP-specific antibody responses is strong at 2 years - currently the longest post-vaccination interval studied - after a single injection. Vaccine dose predicts persistence of seropositivity, though the magnitude of antibody responses at later time points becomes less dose-dependent. Vaccine-related arthritis is a significant predictor of both persistence and magnitude of the antibody response.

Effectiveness of rVSV-ZEBOV vaccination during the 2018-20 Ebola virus disease epidemic in the Democratic Republic of the Congo: a retrospective test-negative study.

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The recombinant vesicular stomatitis virus-Zaire Ebola virus (rVSV-ZEBOV) vaccine is the only WHO

prequalified vaccine recommended for use to respond to outbreaks of Ebola virus (species Zaire ebolavirus) by WHO's Strategic Advisory Group of Experts on Immunization. Despite the vaccine's widespread use during several outbreaks, no real-world effectiveness estimates are currently available in the literature. We conducted a retrospective test-negative analysis to estimate effectiveness of rVSV-ZEBOV vaccination against Ebola virus disease during the 2018-20 epidemic in the Democratic Republic of the Congo, using data on suspected Ebola virus disease cases collected from Ebola treatment centres. Those eligible for inclusion had an available Ebola virus RT-PCR result, available key data, were eligible for vaccination during the outbreak, and had symptom onset aligning with the period in which a ring-vaccination protocol was in use. After imputing missing data, each individual confirmed by RT-PCR to be Ebola virus disease-positive (defined as a case) was matched to one individual negative for Ebola virus disease (control) by sex, age, health zone, and month of symptom onset. Effectiveness was estimated from the odds ratio of being vaccinated (≥ 10 days before symptom onset) versus being unvaccinated among cases and controls, after adjusting for the matching factors. The imputation, matching and effectiveness estimation, was repeated 500 times. 1273 (4.8%) of 26 438 eligible individuals were positive for Ebola virus disease (cases) and 25 165 (95.2%) were negative (controls). 40 (3.1%) cases and 1271 (5.1%) controls were reported as being vaccinated at least 10 days before symptom onset. After selecting individuals who reported exposure to an individual with Ebola virus disease within the 21 days before symptom onset and matching, the analysis datasets comprised a median of 309 cases and 309 controls. 10 days or more after vaccination, the effectiveness of rVSV-ZEBOV against Ebola virus disease was estimated to be 84% (95% credible interval 70-92). This analysis is the first to provide estimates of the real-world effectiveness of the rVSV-ZEBOV vaccine against Ebola virus disease, amid the widespread use of the vaccine during a large Ebola virus disease outbreak. Our findings confirm that rVSV-ZEBOV is highly protective against Ebola virus disease and support its use during outbreaks, even in challenging contexts such as in the eastern Democratic Republic of the Congo. Médecins Sans Frontières. For the French translation of the abstract see Supplementary Materials section.

Knowledge, attitudes, and practices and long-term immune response after rVSVΔG-ZEBOV-GP Ebola vaccination in healthcare workers in high-risk districts in Uganda.

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The rVSVΔG-ZEBOV-GP Ebola vaccine (rVSV-ZEBOV) has been used in response to Ebola disease outbreaks caused by Ebola virus (EBOV). Understanding Ebola knowledge, attitudes, and practices (KAP) and the long-term immune response following rVSV-ZEBOV are critical to inform recommendations on future use. We administered surveys and collected blood samples from healthcare workers (HCWs) from seven Ugandan healthcare facilities. Questionnaires collected information on demographic characteristics and KAP related to Ebola and vaccination. IgG ELISA, virus neutralization, and interferon gamma ELISpot measured immunological responses against EBOV glycoprotein (GP). Overall, 37 % (210/565) of HCWs reported receiving any Ebola vaccination. Knowledge that rVSV-ZEBOV only protects against EBOV was low among vaccinated (32 %; 62/192) and unvaccinated (7 %; 14/200) HCWs. Most vaccinated (91 %; 192/210) and unvaccinated (92 %; 326/355) HCWs wanted to receive a booster or initial dose of rVSV-ZEBOV, respectively. Median time from rVSV-ZEBOV vaccination to sample collection was 37.7 months (IQR: 30.5, 38.3). IgG antibodies against EBOV GP were detected in 95 % (61/64) of HCWs with vaccination cards and in 84 % (162/194) of HCWs who reported receiving a vaccination. Geometric mean titer among seropositive vaccinees was 0.066 IU/mL (95 % CI: 0.058-0.076). As Uganda has experienced outbreaks of Sudan virus and Bundibugyo virus, for which rVSV-ZEBOV does not protect

against, our findings underscore the importance of continued education and risk communication to HCWs on Ebola and other viral hemorrhagic fevers. IgG antibodies against EBOV GP were detected in most vaccinated HCWs in Uganda 2–4 years after vaccination; however, the duration and correlates of protection warrant further investigation.

Characterization of rVSVΔG-ZEBOV-GP glycoproteins using automated capillary western blotting.

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Ebolavirus (EBOV) entry to host cells requires membrane-associated glycoprotein (GP). A recombinant vesicular stomatitis virus vector carrying Zaire Ebola virus glycoprotein (rVSV-ZEBOV) was developed as a vaccine against ebolaviruses. The VSV glycoprotein gene was deleted (rVSVΔG) and ZEBOV glycoprotein (GP) was inserted into the deleted VSV glycoprotein open reading frame (ORF) resulting in a live, replication-competent vector (rVSVΔG-ZEBOV-GP). Automated capillary westerns were used to characterize the rVSVΔG-ZEBOV-GP vaccine (ERVEBO®) manufacturing process with regards to glycoprotein (GP) structure and variants. The method shows a unique electropherogram profile for each process step which could be used to monitor process robustness. rVSVΔG-ZEBOV-GP encodes GP (GP1-GP2), secreted GP (sGP), and small secreted GP (ssGP) variants. Furthermore, a TACE-like activity was observed indirectly by detecting soluble GP2Δ after virus precipitation by ultracentrifugation. Capillary western blotting techniques can guide process development by evaluating process steps such as enzyme treatment. In addition, the technique can assess GP stability and process lot-to-lot consistency. Finally, capillary western-based technology was used to identify a unique biochemical profile of the rVSVΔG-ZEBOV-GP vaccine strain in final product. Virion membrane-bound GP1-GP2 is critical to vaccine-elicited protection by providing both neutralizing antibodies and T-cell response.

Polyclonal and convergent antibody response to Ebola virus vaccine rVSV-ZEBOV.

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Recombinant vesicular stomatitis virus-Zaire Ebola virus (rVSV-ZEBOV) is the most advanced Ebola virus vaccine candidate and is currently being used to combat the outbreak of Ebola virus disease (EVD) in the Democratic Republic of the Congo (DRC). Here we examine the humoral immune response in a subset of human volunteers enrolled in a phase 1 rVSV-ZEBOV vaccination trial by performing comprehensive single B cell and electron microscopy structure analyses. Four studied vaccinees show polyclonal, yet reproducible and convergent B cell responses with shared sequence characteristics. EBOV-targeting antibodies cross-react with other Ebolavirus species, and detailed epitope mapping revealed overlapping target epitopes with antibodies isolated from EVD survivors. Moreover, in all vaccinees, we detected highly potent EBOV-neutralizing antibodies with activities comparable or superior to the monoclonal antibodies currently used in clinical trials. These include antibodies combining the IGHV3-15/IGLV1-40 immunoglobulin gene segments that were identified in all investigated individuals. Our findings will help to evaluate and direct current and future vaccination strategies and offer opportunities for novel EVD therapies.